Real-Time State of Charge Estimation of Lithium-Ion Batteries Using Optimized Random Forest Regression Algorithm

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***Abstract*—This paper presents an improved machine learning approach for the accurate and robust state of charge (SOC) in electric vehicle (EV) batteries using differential search optimized random forest regression (RFR) algorithm. The precise SOC estimation confirms the safety and reliability of EV. Nevertheless, SOC is influenced by numerous factors which cannot be measured directly. RFR is suitable for real-time SOC estimation due to its robustness to noise, overfitting issues and capacity to work with huge datasets. However, proper selection of RFR architecture and hyper-parameters combination remains a key issue to be explored. Hence, a differential search algorithm (DSA) is employed to search for the optimal values of trees and leaves in the RFR algorithm. DSA optimized RFR eliminates the utilization of the filter in data pre-processing steps and does not require a detailed understanding and knowledge about battery chemistry, rather only needs sensors to monitor battery voltage and current. The developed approach is validated at room temperature using two types of lithium-ion batteries under a pulse discharge test. In addition, the proposed model is verified under varying temperature settings under EV drive cycles. The experimental results demonstrate that the DSA optimized RFR algorithm achieves RMSE of 0.382% in the HPPC test using LiNMC battery. Besides, the proposed method obtains satisfactory outcomes in EV drive cycles, estimating MAE of 0.193% and 0.346% in DST and FUDS cycles, respectively, at 25°C.**

***Index Terms*—Differential search algorithm, electric vehicle, lithium-ion battery, random forest regression, state of charge.**

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I. Introduction

THE electric vehicle (EV) and related battery technology have seen significant advancement in addressing the carbon emission issues [1]. The effectiveness of EV largely depends on precise assessment, enhanced capabilities, diagnosis and prognostic of the battery storage system [2]. At present, lithium-ion batteries outperform other batteries in EV applications in achieving a long lifespan, low self-discharge rates, high voltage and high energy density [3]. One of the key parameters to assess the performance of a battery is the state of charge (SOC) which indicates the level of charge in a battery cell to drive an EV [4]. The battery SOC depends on various parameters related to electrochemical reactions, material degradation and aging cycles [5]. The estimation of SOC is a complex task due to several key issues such as inappropriate battery model, high mathematical computation, poor robustness and slow estimation speed caused by noise and temperature deviations [6]. Therefore, an improved algorithm for SOC estimation is essential to attain safe, reliable and stable working of lithium-ion batteries in EVs.

The estimation of SOC is classified into two groups: model-based approaches and data-driven approaches [7]. In a model-based approach, set rules and the mathematical model are developed with in-depth domain knowledge of the battery with a reasonable volume of data [8]. In line with that, battery anode and cathode features, solid-phase diffusion, electrolytic reaction, polarization are needed to be studied [9]. Besides, the domain expert and time to execute the experiments need to be explored [10]. Hu *et al.* [11] used a fractional-order calculus-based Kalman filter (KF) algorithm to estimate SOC for the lithium-ion battery management system. The model is verified by experimental tests and results indicated SOC error under 1% with the presence of noise, initial deviation, and disturbance. Deng *et al.* [12] employed the joint estimation method based on sigma-point Kalman filtering (SPKF) to determine the lithium-ion model parameters and SOC. The proposed model is verified by different cases and the results show the absolute error of SOC and voltage under 1.5% and 2.1 mV, respectively. Chen *et al.* [13] proposed voltage-based state of charge (SOC) estimation using the particle filter. The static and dynamic tests are employed to determine the model parameters and develop the relationship between voltage-based SOC and capacity. The results demonstrated that the voltage-based SOC has lower error

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under different temperature and current conditions. Xia *et al.* [14] introduced a strong tracking H-infinity filter (STHF) to estimate SOC for lithium-ion batteries. The proposed model is validated by a battery test bench at different temperatures. The results show high accuracy and robustness under different initial SOC errors and measured noise. Furthermore, STHF delivered better SOC estimation results compared to KF.

In contrast, data-driven methods have become increasingly popular in SOC estimation since they do not require set prescribed rules and mathematical models. In recent times, data-driven methods have received huge attention due to the accessibility of sufficient data and high computation power [15]. The data-driven approaches achieve high efficiency and performance in SOC estimation due to their strong computation capability under various constraints such as noise, varying temperatures and aging effects [16]. Although data-driven methods have presented promising outcomes towards the enhancement of SOC accuracy, they exhibit certain limitations. For instance, radial basis function neural networks [17], extreme learning machine [18] deliver reasonable SOC outcomes but their accuracy is affected by inaccurate hyperparameters settings. The Gaussian process regression [19] and support vector machine [20] can address noise, and data overfitting problems, but they lack accuracy under highly non-linear systems. The time-delay neural network [21], recurrent nonlinear autoregressive with exogenous input neural network [22] are good in obtaining accurate SOC results under temperature and aging cycles, but they need a careful selection of the past information. Deep learning techniques such as long short-term memory [16], gated recurrent unit [23] provide excellent results under diverse load conditions but require high computational power with a large amount of data. The optimization techniques such as particle swarm optimization [24], genetic algorithm [25], lighting search algorithm [26] can be combined with data-driven methods but they suffer from drawbacks like deviation of the accuracy due to inappropriate parameter settings, poor exploration and exploitation capability and local minima trap.

Various factors such as training methodology, input features, hyperparameter, activation function etc. determine the SOC effectiveness using data-driven methods. The hyperparameters like hidden neurons, hidden layers, weights, bias, iterations, sampling intervals etc. are crucial in developing the data-driven method structure [27]. The computational complexity of data-driven methods is dropped with the appropriate selection of the hyperparameters leading to a decrease in data underfitting and overfitting problems [6]. Generally, the hyperparameters and functions to train the data-driven methods are chosen using the inefficient trial-and-error approach, leading to the loss of time and human energy [26]. Hence, an appropriate framework and hyperparameter tuning of data-driven methods toward accurate SOC estimation need careful investigation.

In this paper, a data-driven optimization method for accurate SOC estimation is presented using random forest regression (RFR) algorithm integrated with a differential search algorithm (DSA). RFR is regarded as a subclass of machine learning that is effective against noise and over-fitting issues. The capability of the DSA based RFR is enhanced by selecting the optimal hyperparameters, thereby increasing SOC estimation accuracy. The major contributions of this study are highlighted below:

- The battery data like the voltage, current, and temperature are obtained from sensors in the proposed DSA based RFR algorithm which excludes the utilization of the filter in data pre-processing steps.
- The hyperparameters of RFR including tress and leaves are updated by themselves without taking into consideration the deeper understanding and knowledge used in model-based SOC estimation.
- The mathematical model and complexity to construct robust rules to capture battery behavior are eliminated as DSA based RFR algorithm to estimate SOC does not require the battery model.
- DSA based RFR algorithm increases the SOC estimation accuracy by choosing the best values of hyperparameter without any complex and time-consuming trial and error method.
- The presented approach can be applied to various chemistry of EV batteries as well as EV drive cycles under varying temperature conditions.

The remaining paper is structured into five sections. Section two narrates the methodological framework of the SOC estimation model using DSA based RFR algorithm. Section three outlines the test bench platform, battery experimental tests, and data preparation. The experimental validations are highlighted in section four. The concluding remarks are presented in section five.

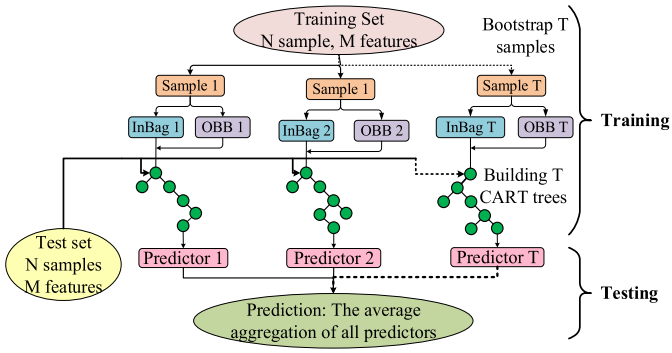


Fig. 1. Flow diagram of RFR algorithm structure.

II. Hybrid SOC Algorithm Framework Based on Optimized Random Forest Regression

A. Random Forest Regression Algorithm

RFR algorithm is based on a set of predictors which depend on trees in the forests through the random values of each tree. The modeling of RFR is observed by extracting input datasets from a group and further dividing it into a random order. The superiority in the performance of the RFR algorithm is due to efficiently handling large datasets, fast operational speed, and does not overfit as a predictor. [28]. The flow diagram of the RFR algorithm is shown in Fig. 1.

The RFR regression is developed using the p dimension input vector of $X = x_1, x_2, \ldots\ldots, x_p$ to build the forest. A set of K tress $\{T_1(x), T_2(x), \ldots, T_k(x)\}$ is used inside the forest. The

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actual output value is determined by each tree, represented as $\hat{Y}_1 = T_1(X), \ldots, \hat{Y}_m = T_1(X)$, where $m = 1, \ldots, K$. The outcome of RFR is evaluated by estimating the average of all trees predictors, as expressed in the following equation.

$$Predict_{RF}(X) = \frac{1}{k}\sum_{k=1}^{K}\widehat{Y_N}(X) \tag{1}$$

The training dataset $D = D_1, D_2, \ldots, D_n, = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ is drawn independently from input and output where $x_i, i = 1, \ldots, n$ denotes the input vector training dataset and $y_i, i = 1, \ldots, n$ expresses the output vector training dataset. The training procedures for the growth of tree in the forest are explained as follows [28].

- *Step 1:* A bootstrap sample is drawn for each regression tree through the available training dataset. A bootstrap sample is a random sample conducted with replacement. A different subset of the dataset is employed to improve the tree model for each bootstrap sample. The Out-of-bag (OBB) samples are formed by leaving one-third of the dataset. A total of two-thirds sample is available of the new training sample. OBB is the process of neglecting values from each bootstrap sample. OBB data plays a key part in tree development and is checked with the estimated values at each step.
- *Step 2:* At each node of the regression tree in the bootstrap sample, the best split is chosen among the randomly selected subset. The node in each tree plays a vital role in the correction of changing parameters of the algorithm.
- *Step 3:* Each tree is designed to the largest extent possible without pruning.
- *Step 4:* The predictions are evaluated by placing each OBB observations of the test data for each tree. The mean value of predictions of the total regression trees is calculated using (1).

The minimization of OBB is observed and further accuracy and error rate is calculated. One of the important features of RFR is OBB error. As mentioned earlier, the development of each tress is dependent on the bootstrap sample consisting of nearly two-thirds of the training data. The remaining data (OBB) of the training data is utilized for testing. The MSE of OBB error is evaluated by calculating the deviation between predicted and reference values, as shown the following equation,

$$MSE \approx MSE^{OBB} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{Y}(X_i) - Y_i\right)^2 \tag{2}$$

where $\hat{Y}(X_i)$ denotes the predicted output, Y_i represents the observed output and *n* is the total number of samples.

B. Differential Search Algorithm

DSA is a population-based optimization algorithm that uses high-level stochastic method to simulate the Brownian like random movement of search organisms [29]. In this algorithm, the individual search organisms from a bigger population known as a superorganism and start moving altogether to discover efficient habitats. The search direction of a superorganism is influenced by the donors. The donors are assumed to be in a fertile area for the discovery of stopover sites. The entire search process of DSA is classified into five major steps. They are described as follows [30].

1) Initialization: The generation of the population (superorganism) in DSA depends upon uniform distribution within the boundary constraints. Let *D* represents optimization problem dimension and *N* is population size respectively, then initialization of each individual of the population (pop) is given as under,

$$\text{Initial population}, pop_{n,d} = low_d + randx(up_d - -low_d) \tag{3}$$

$$X_{pop} = pop \tag{4}$$

where n € {1,2,3, … N} and d € {1,2,3, … D}. Here *r and* is the function for random number generation based on the uniform distribution (0∼1). The search space is defined by the boundary constraints which are row vectors as follow,

$$low_d = [low_1 low_2 \cdots ..low_D] \tag{5}$$

$$up_d = [up_1 up_2 \cdots up_D] \tag{6}$$

The fitness value for each individual is determined by evaluating the objective function (f) as follow,

$$Objective function, y_{pop} = f(x_{pop}) \tag{7}$$

2) Stopover Site Discovery: In DSA, superorganism (xpop) is assumed to discover the global minima point of the problem. The successful migration of superorganism depends on the mechanism of stopover site discovery. The search direction of a superorganism is towards the fertile area of donors. The donors are generated by permuting the main superorganism, as denoted in (8). The scale factor F is generated based on the gamma random function as shown in (9). In DSA, the discovery of the stopover site is defined by (10).

$$x_{donor} = permuting(x_{pop}) \tag{8}$$

$$\text{Scale factor}, F = randg(2.rand).(rand - rand) \tag{9}$$

$$\text{Trial stopover site}, x_{stopover} = x_{pop} + F.x(x_{donor} - x_{pop}) \tag{10}$$

3) Selection of Search Participants: The selected number of participants involve in the search of stopover sites unlike other algorithms. In this regard, the participants are selected randomly by generating a selection index (r). It is worth mentioning that there are only two control parameters in DSA. Both control parameters are used to randomize the selection process of search participants. As they do not have any direct influence over organism's movement, the choice of control parameters does not have direct influence over the optimization solution. The index, r is a binary matrix and used to select the search participants in the stopover site discovery. It is generated based on the randomization technique. The selection of search particles is expressed as follows,

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Selected search participants,

$$\mathrm{x}_{\mathrm{stopover_selected}} = \begin{array}{c} Xstopover; n.d \ < if \ rn, d \ = \ 1 \\ or \\ Xstopover; n.d \ < if \ n, d \ = \ 0 \end{array} \quad (11)$$

4) Boundary Checking: During the movement of superorganism, some of the participants may go beyond the search space limits (boundaries). Those participants are redirected to within the search space based on (12) as follow,

$$x_{stopover_final} = low_d + rand\, x\left(up_d - -low_d\right), \quad if \begin{cases} Xstopover_selected; n.d \ < lowd \\ or \\ Xstopover_selected; n.d \ < upd \end{cases} \quad (12)$$

The success in the stopover site discovery is updated by the main population. The participants that discover the stopover sites are laid over that location and continue their next journey from that location.

Index for successful discovery,

$$k_{index} = y_{\ stopover_final} > Y_{pop} \quad (13)$$

Updating main population

$$x_{pop}\left(k_{index}\right) = x_{stopover_final}(k_{index}) \quad (14)$$

C. Parameter Optimization, Objective Function and Constraints

DSA utilizes the objective function to reach the global optimum solution leading to lower SOC error by exploring the suitable hyperparameters in RFR. The lowest error rate of the objective function is obtained by DSA through the iteration which corresponds to optimum values of the trees and leaves of RFR. The study aims to achieve the objective function by reaching the minimum root means square error (RMSE) that can be written as,

$$Objective function = \min\left(RMSE\right) \quad (15)$$

The optimization constraints are formulated using the upper and lower number of the trees and leaves of RFR algorithm. The accuracy of DSA can deviate if the hyperparameter population is outside the boundary. Hence, for every iteration, the values of the hyperparameters are checked. The values will be generated and updated again if the values stay outside the boundary conditions. The limit of hyperparameters constraints can be expressed as follows,

$$X_{i,j}^{k-1} < X_{i,j}^{k} < X_{i,j}^{k+1} \quad (16)$$

where $X_{i,j}^{k-1}$ and $X_{i,j}^{k+1}$ denote the lower and upper value of $X_{i,j}^{k}$

D. Execution Process of DSA Based RFR for SOC Estimation

The execution process for DSA based RFR algorithm initiates with monitoring current and voltage of the battery. Furthermore, the appropriate values of trees and leaves based on objective function are obtained by employing the DSA algorithm. Finally, the SOC prediction accuracy is obtained using various performance metrics. The methodological framework of the proposed algorithm is depicted in Fig 2.

In stage I, the test bench model is designed to conduct the HPPC battery experimental tests. Besides, various EV battery data like voltage, current and temperature are obtained from the CALCE dataset [26]. Furthermore, pre-processing and data normalization are carried out to improve the training speed. Finally, the data division into training and testing is executed using the cross-validation method.

In stage II, various hyperparameters such as the number of iterations, size of the population, input dimension, objective function, and optimization constraints are selected for executing the DSA. The steps to implement DSA based RFR are summarized as follows:

- *Step 1:* The optimization is initialized by declaring a base setting for DSA. Initial population (xpop) of DSA is generated using (3-4) The boundary limits (low, up) of optimizing parameters are set using (5-6). In the proposed methodology, the population size is assigned as 50 while the iteration number is assigned as 500. This step also defines the limit of trees and leaves is between '1-500' and '1-5', respectively.
- *Step 2:* The objective function is formulated using (7).
- *Step 3:* Donors (xdonor) and scale factor (F) are generated using. (8) and (9), respectively. Trail stopover sites (xstopover) are estimated following (10).
- *Step 4:* The selection index (r) is generated to choose search participants in DSA. Equation (11) is used to keep the stopover sites (xstopover_selected) for selected search participants.
- *Step 5:* The search boundaries are checked for stopover sites. If any of them cross the limits, equation (12) is used to update the final stopover sites.
- *Step 6:* Calculation of objective function by repeating Step 2.
- *Step 7:* If any fruitful discovery of stopover site occurs then the superorganism migrates using (12) and (13).
- *Step 8:* The iteration is kept continue until the maximum generation of optimization iteration is met.
- *Step 9:* The results of optimization is displayed when it finishes the optimization.

In stage III, the verification of the proposed algorithm is observed through HPCC experiments is calculated by the statistical error rate as indicated in (18)-(20). Furthermore, the robustness of the DSA based RFR algorithm is checked at varying temperatures under EV drive cycles.

III. Battery Experiments and Data Preparation

A. Battery Test Bench Development

This research uses two types lithium-ion batteries that are commonly used in EVs. The first battery cell is produced by Panasonic NCR18650B which has a capacity of 3200 mAh while the second cell is manufactured by Samsung ICR18650-26F which is rated as 2600 mAh [27]. The graphite acts as the negative electrode while $LiNiCoAlO_2$ (LiNCA) and $LiNiMnCoO_2$ (LiNMC) act as positive electrodes for NCR18650B and ICR18650-26F battery, respectively. The LiNMC and LiNCA batteries are dominant in EVs due to their extended lifespan while the latter offers high specific energy. Table I presents

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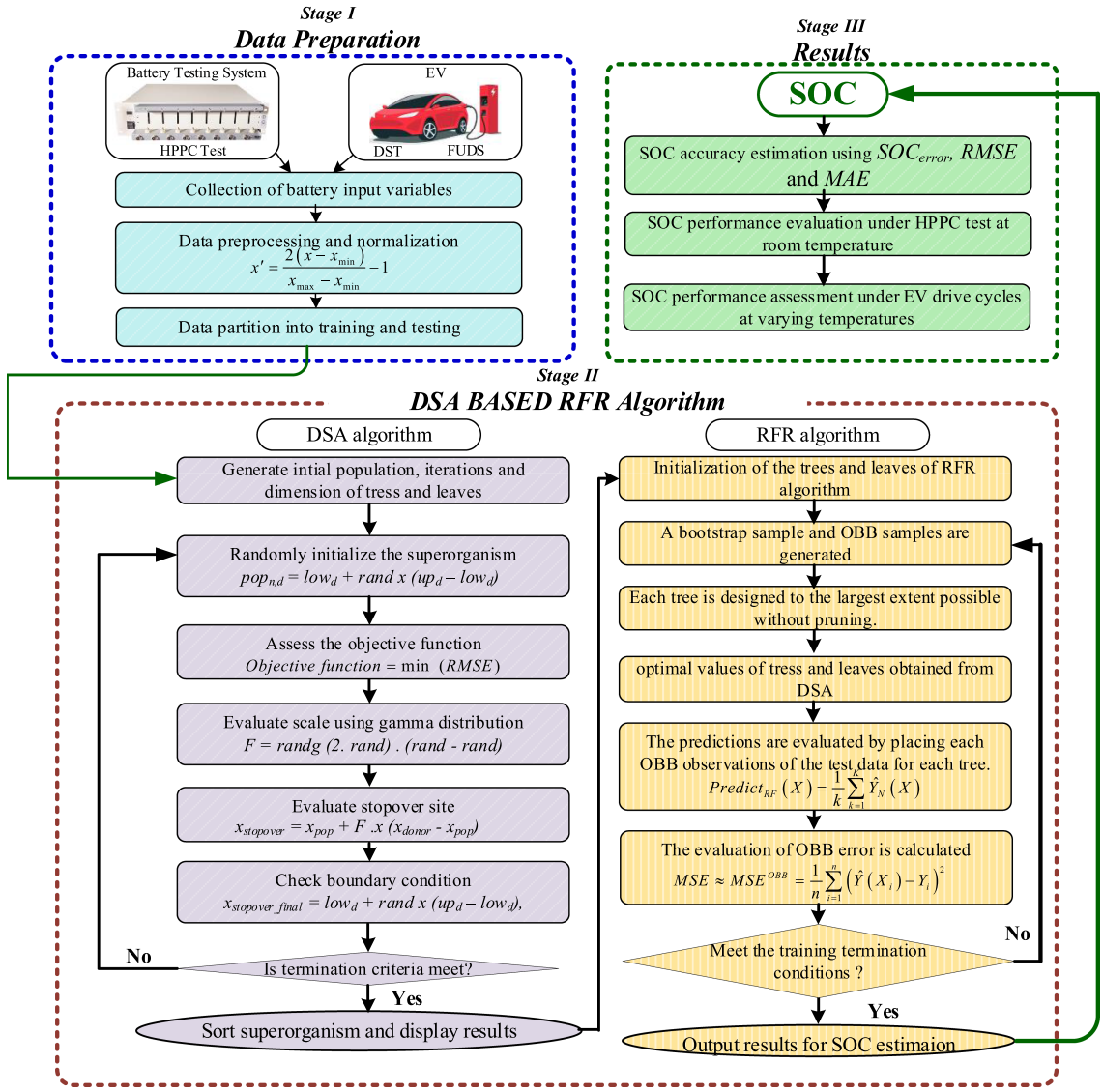


Fig. 2. Flow diagram for SOC estimation based on DSA optimized RFR algorithm.

TABLE I
LITHIUM-ION BATTERY SPECIFICATIONS

Parameters	$LiNiCoAlO_2$	$LiNiMnCoO_2$
Nominal capacity (Ah)	3.2 Ah	2.6 Ah
Nominal Voltage (V)	3.6	3.7
Min/Max voltage (V)	2.5/4.2	2.75/4.2
Specific Energy (Wh/kg)	200-260	150-220
Charging current (mA)	1625	1300
Charging method	CC-CV	CC-CV
Weight (g)	48.5	47.0
Lifespan (cycles)	500	1000-2000
Thermal runaway (temperature °C)	150	210

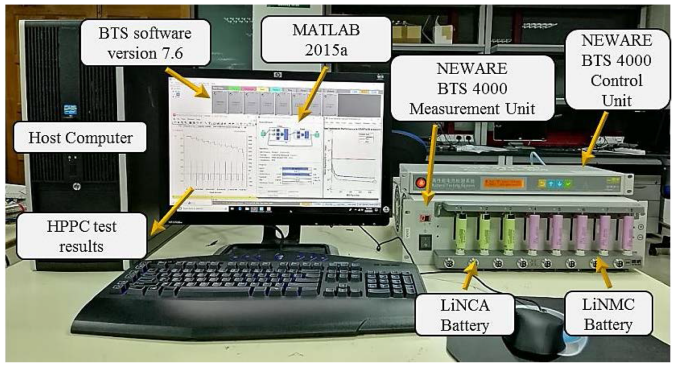


Fig. 3. Test bench model configuration for SOC estimation.

the specification of various parameters for $LiNiCoAlO_2$ and $LiNiMnCoO_2$ batteries.

An experimental test bench model is created to monitor battery current, voltage and capacity as well as estimate SOC. Both hardware and software sections are used to establish the test bench, as presented in Fig. 3. The hardware model consists of the measurement and control unit. The measurement unit is configured using LiNCA, LiNMC batteries and NEWARE BTS-4000. The 8 channels of NEWARE BTS-4000 are used to record the battery data in each second with an accuracy of $\pm$ 0.05%. The

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range of voltage and current for the hardware measurement unit is 5 V and 6 A, respectively. The control unit prevents the battery from overcharging and over-discharging problems by keeping the voltage under cut-off range. The software section consists of BTS software version 7.6 and MATLAB 2015a which are installed in the host computer. Firstly, the BTS software is used to enter the settings of HPPC experiment and send instructions to the hardware section. Accordingly, BTS hardware follows the commands and saves data in the host computer. Finally, the host computer uses the data to execute the algorithm using MATLAB 2015a and estimate SOC.

B. Experimental Tests

The hybrid pulse power characterization (HPPC) test is carried out to evaluate SOC at room temperature. Before the beginning of the experimental work, the battery is fully charged and SOC of the battery is observed at 100%. The steps of each experimental test are elaborated using the Samsung ICR18650-26F battery. The HPPC test performs the charging and discharging of the current pulses in an assigned manner which is presented as under.

i. The constant current-constant voltage (CC-CV) technique is utilized to charge the battery voltage till the current of the battery declines to 0.13 A (0.05 C).
ii. A rest period for 1 hour is executed.
iii. The battery is discharged with 1.3 A (0.5 C) for 10 seconds.
iv. The battery is kept at rest mode for 3 minutes.
v. The battery is charged with 1.3 A (0.5 C) for 10 seconds.
vi. The battery is rested for 3 minutes.
vii. The SOC capacity is reduced by 10% when the battery is discharged with 0.65 A for 24 minutes.
viii. When the lower cut-off voltage is reached, the experiment is terminated otherwise, step 3 is initiated again.

C. Data Pre-Processing, Training and Testing

The data obtained from the experiment is arranged into two sets i.e., training and testing. The cross-validation method with 70-30 ratio is utilized to obtain the training and testing dataset [26]. The training of the proposed algorithm is observed under offline mode while the SOC is monitored during online mode. To enhance the training accuracy as well as speed, the normalization of the dataset is executed within the boundary of [-1, 1], as shown in (17),

$$x' = \frac{2\left(x - x_{\min}\right)}{x_{\max} - x_{\min}} - 1 \tag{17}$$

where x_{max} and x_{min} represents the maximum and minimum value of variable x, respectively. The host computer has the configuration of core i9, 3.70 GHz and 32 GB random access memory (RAM). The HPPC load profile for LiNCA and LiNMC batteries is presented in Fig. 4. In this paper, the positive current relates to charging while the negative current relates to the discharging process. The SOC of the battery increases proportionally with voltage thus establishing the relationship between SOC and voltage as shown in Fig 5.

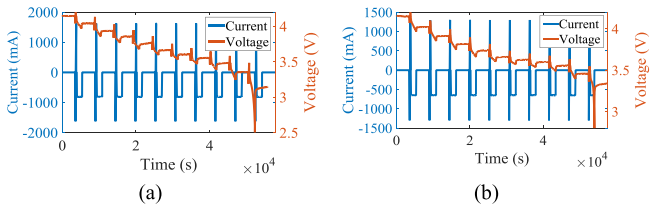


Fig. 4. Current and voltage under HPPC test profile (a) LiNCA battery and (b) LiNMC battery.

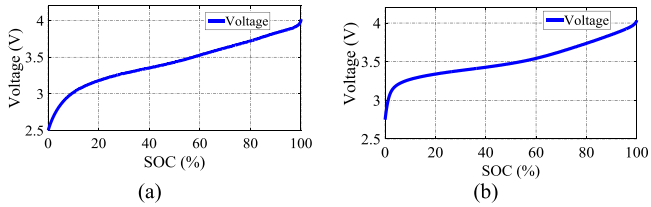


Fig. 5. Battery voltage with variation of SOC (a) LiNCA battery and (b) LiNMC battery.

D. SOC Performance Indicators

The performance of the proposed algorithm is evaluated with regards to the reference SOC values obtained by adjusting the current sensor in the Columb counting method. The performance metrics such as $\mathrm{SOC_{error}}$, RMSE and mean absolute error (MAE) in predicting the SOC is expressed below,

$$SOCerror = SOC_{a_i} - SOC_{es_i} \tag{18}$$

$$RMSE = \sqrt{\frac{1}{N}\sum\nolimits_{i=1}^{N}\left(SOC_{a_i} - SOC_{es_i}\right)^2} \tag{19}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|\left(SOC_{a_i} - SOC_{es_i}\right)\right| \tag{20}$$

where SOC_a refers to the reference SOC, SOC_{es} refers to the predicted SOC by the proposed algorithm and N refers to the number of data size.

IV. Experiments and Results Analysis

A. SOC Accuracy at Room Temperature

The adaptability and accuracy of the DSA based RFR algorithm for SOC estimation are verified under HPPC discharge current load profile at room temperature [27]. To check the proposed model accuracy, a comparative study between the proposed method and commonly used machine learning and deep learning algorithms including ELM, SVM and LSTM is conducted. ELM is formed using three layers and implementation is carried out by assigning the hidden layer biases and input weights randomly. ELM exhibits fast training execution, good learning skill and high accuracy. However, the accuracy of ELM depends on the proper selection of hidden layer neurons [18]. The SVM is employed for data analysis related to classification and regression. The data characterization in SVM is defined by the hyperplanes in high dimensional space. Although SVM delivers satisfactory performance in non-linear and high dimension models, it has high complex computation [31]. The LSTM addresses the issue of exploding or vanishing gradient

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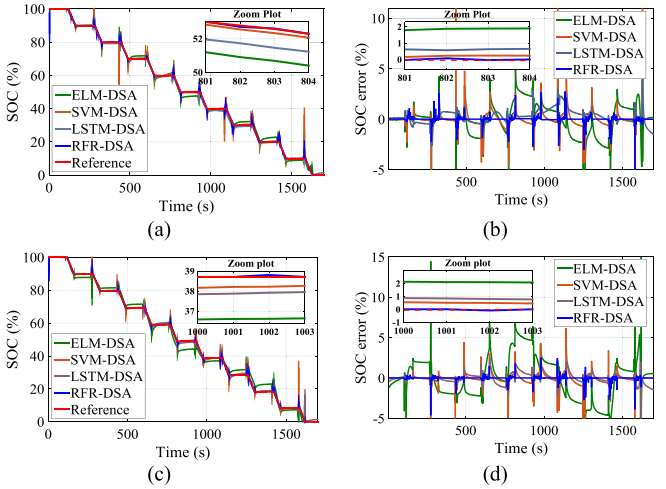


Fig. 6. SOC estimation accuracy at room temperature (a-b) LiNCA battery (c-d) LiNMC battery.

TABLE II
SOC ACCURACY AT ROOM TEMPERATURES IN HPPC TEST

Method	Battery	RMSE (%)	MAE (%)	SOC error (%)
ELM-DSA	LiNCA	1.563	1.143	[-8.56, 5.78]
	LiNMC	2.797	0.785	[-8.78, 14.48]
SVM-DSA	LiNCA	1.327	0.437	[-6.64, 5.77]
	LiNMC	1.507	0.423	[-7.58, 6.16]
LSTM-DSA	LiNCA	0.774	0.552	[-2.72, 7.56]
	LiNMC	0.747	0.483	[-7.38, 2.48]
RFR-DSA	LiNCA	0.508	0.118	[-4.86, 2.56]
	LiNMC	0.382	0.086	[-4.62, 2.42]

in the conventional recurrent neural network and applies the memory units instead of ordinarily hidden nodes to capture the long-term dependencies. Nonetheless, LSTM has complex training execution and needs an expensive device to accelerate training [32]. The hyperparameters including hidden neurons in ELM, kernel parameters in SVM, learning rate in LSTM are optimized by DSA to conduct a fair comparative analysis. In case battery experimental tests, the number of trees for LiNCA and LiNMC battery experimental tests are calculated as 425 and 387, respectively while it is 4 and 3 for leaves, respectively. It is noticed from Fig. 6 that reference SOC and SOC by DSA based RFR algorithm are placed close to each other. Nonetheless, SOC values determined by DSA based ELM, DSA based SVM and DSA based LSTM are moving away from the reference SOC values. The results indicate that DSA based RFR algorithm has satisfactory results, having SOC error under 5% for both LiNCA and LiNMC batteries.

The different statistical error terms also confirm the dominance of DSA based RFR algorithm over DSA optimized machine learning and deep learning approaches as outlined in Table II. For example, RMSE in LiNCA battery is achieved to be 0.508% which is reduced by 67.49%, 61.71% and 34.37%, as compared to DSA based ELM, DSA based SVM and DSA based LSTM methods, respectively. Besides, the proposed model has MAE of 0.086% in LiNMC battery which is a drop of 89.04%, 79.67%, and 82.14% from the DSA based ELM, DSA based SVM and DSA based LSTM methods, respectively.

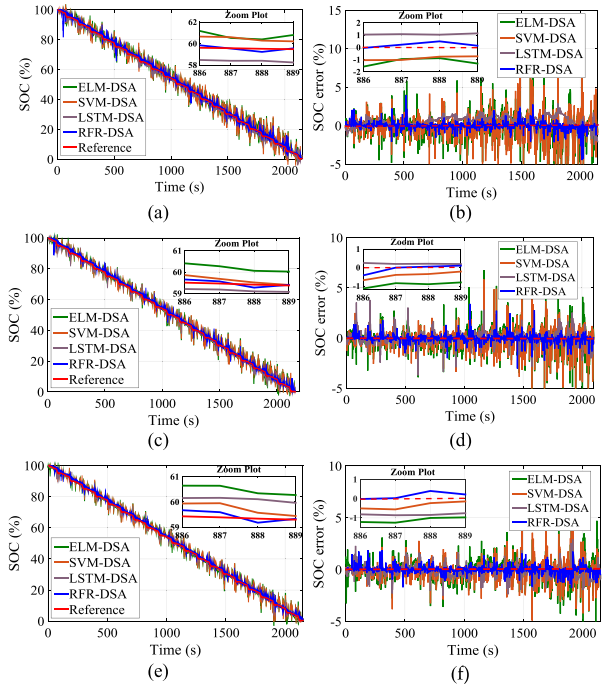


Fig. 7. SOC estimation accuracy under DST drive cycle. (a-b) 0°C. (c-d) 25°C. (e-f) 45°C.

TABLE III
SOC ACCURACY AT VARYING TEMPERATURES IN DST DRIVE CYCLE

Method	Battery	RMSE (%)	MAE (%)	SOC error (%)
ELM-DSA	0°C	1.698	1.167	[-10.35, 12.5]
	25°C	1.132	0.784	[-6.27, 6.78]
	45°C	0.943	0.671	[-5.67, 9.24]
SVM-DSA	0°C	1.504	1.008	[-11.26, 8.56]
	25°C	1.078	0.685	[-7.41, 6.01]
	45°C	0.784	0.524	[-4.91, 6.75]
LSTM-DSA	0°C	0.903	0.765	[-3.73, 3.81]
	25°C	0.662	0.472	[-2.51, 2.89]
	45°C	0.574	0.427	[-2.48, 2.92]
RFR-DSA	0°C	0.658	0.249	[-4.46, 3.85]
	25°C	0.425	0.193	[-3.11, 3.42]
	45°C	0.312	0.183	[-1.71, 1.87]

B. SOC Accuracy at Varying Temperatures

The performance of SOC estimation is checked under varying temperatures under Dynamic Stress Test (DST) as presented in Fig. 7. The SOC estimated by DSA based RFR algorithm is very adjacent to the reference SOC which demonstrates high accuracy. In addition, the DSA based RFR is outstanding in delivering low SOC error, RMSE, and MAE, as depicted in Table III. For instance, at 25°C, DSA based RFR computes RMSE and MAE to be 0.425% and 0.193%, respectively which is lower than DSA based ELM, DSA based SVM and DSA based LSTM methods. Moreover, at 45°C, about 66.91%, 60.2% and 45.64% reductions are noted in DSA based RFR methods compared to DSA based ELM, DSA based SVM and DSA based LSTM methods, respectively while calculating RMSE. In all temperature settings, SOC error in DSA based RFR stays under 5%. In contrast, SOC estimated by DSA based RFR under Federal Urban Driving Cycles (FUDS) is satisfactory and the

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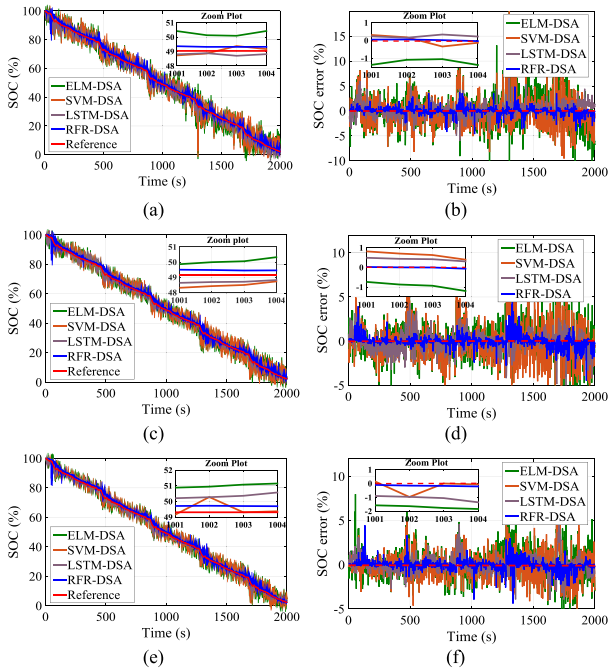


Fig. 8. SOC estimation accuracy under FUDS drive cycle. (a-b) 0°C. (c-d) 25°C. (e-f) 45°C.

TABLE IV
SOC ACCURACY AT VARYING TEMPERATURES IN FUDS DRIVE CYCLE

Method	Battery	RMSE (%)	MAE (%)	SOC error (%)
ELM-DSA	0°C	3.434	2.537	[-12.76, 13.17]
	25°C	2.187	1.629	[-8.62, 10.14]
	45°C	1.732	1.287	[-6.12, 9.16]
SVM-DSA	0°C	2.753	1.896	[-11.86, 18.58]
	25°C	1.951	1.382	[-6.52, 10.62]
	45°C	1.423	0.994	[-5.69, 9.38]
LSTM-DSA	0°C	1.369	0.978	[-6.32, 6.57]
	25°C	1.167	0.879	[-6.13, 4.29]
	45°C	0.967	0.656	[-3.99, 5.16]
RFR-DSA	0°C	0.912	0.424	[-4.65, 4.87]
	25°C	0.823	0.346	[-4.48, 4.78]
	45°C	0.704	0.307	[-4.39, 4.45]

values are located near to reference SOC, as shown in Fig. 8. Nevertheless, SOC estimated by DSA based ELM, DSA based SVM and DSA based LSTM is fluctuating and moving apart from reference SOC. In case of EV drive cycles, the number of trees and leaves are decreasing with the rise of temperature. For instance, in DST drive cycle, the number of trees is determined as 476, 459 and 436 at 0°C, 25°C and 45°C, respectively. This similar kind of results is also found in FUDS drive cycle under different temperatures.

The RMSE and MAE values in Table IV also ensure the superiority of DSA based RFR performance over DSA based ELM, DSA based SVM and DSA based LSTM algorithms. For instance, in case of RMSE at 0°C, there is a decrease of 73.44%, 66.87% and 33.38% in DSA based RFR method in comparison to DSA based ELM, DSA based SVM and DSA based LSTM methods, respectively. Besides, MAE at 25°C in the proposed method is decreased by 78.76%, 74.96% and 60.64% as compared to DSA based ELM, DSA based SVM and DSA based LSTM methods, respectively. Furthermore, at 45°C, DSA based RFR computes maximum SOC error to be 4.45% while DSA based ELM, DSA based SVM and DSA based LSTM methods have a maximum SOC error of 9.16%, 9.38%, 5.16%, respectively. In summary, the results obtained from experiments and EV drive cycles prove that the DSA based RFR is robust and delivers excellent solutions in SOC estimation under diverse loads and temperature conditions.

The temperature rise has a significant impact on battery SOC estimation. The increase in battery temperature leads to an increase capacity of the battery. A rise of the electrolyte activity as well as reduction of viscosity are caused by the temperature acceleration which may support the migration impact and ion diffusion [33]. In this study, it is also observed that the increase in temperature causes a decrease in SOC error, RMSE and MAE. For instance, the proposed DSA based RFR model estimates RMSE of 0.685% in DST drive cycles at 0°C. However, RMSE is reduced to 0.425% and 0.312% at 25°C and 45°C, respectively. Similarly, in FUDS drive cycle, MAE is obtained to be 0.424% at 0°C while it is decreased to 0.346% and 0.307%, at 25°C and 45°C, respectively.

V. CONCLUSION

The overall contribution of this research is the development of an advanced SOC estimation model using DSA optimized RFR algorithm. The proposed algorithm elevates the SOC estimation accuracy by finding the appropriate value of trees and leaves of RFR without considering an exhaustive trial and error approach which needs a considerable duration and human expertise. The performance of DSA optimized GSA is verified through experiments using two different types of lithium-ion cells including LiNCA and LiNMC at room temperature. The proposed algorithm achieves high SOC accuracy having RMSE of 0.508% and 0.382% in LiNCA and LiNMC battery, respectively in HPPC test. The robustness of the proposed algorithm is further validated at varying temperature conditions. The results are excellent in terms of RMSE and MAE indicating below 1% and 0.5%, respectively in DST and FUDS cycles in varying temperatures. The proposed algorithm is also dominant to ELM, SVM and LSTM with regard to high accuracy, adaptability and robustness. In all test conditions and temperatures, SOC error is below 5%. The future works include the verification SOC estimation accuracy under the aging cycle test. Besides, the proposed algorithm can be validated through a hardware-in-the-loop test in real-time.

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